

HIDDEN GEMS

A YouTube Discovery Platform Initiative

Use Cases

Hidden Gems — Niche Creator Discovery Platform

Use Case Diagram & Detailed Specifications

Project	Hidden Gems Discovery Layer (YouTube)
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Revision History

Version	Date	Author	Description of Change
0.1	2026-07-03	Business Analyst	Initial actor/use-case identification
1.0	2026-07-14	Business Analyst	Baselined detailed specifications

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1. Actors

Actor	Description
Creator	A mid-tier YouTube creator (1K–500K subscribers) who publishes content and wants to be discovered by relevant viewers.
Viewer	A YouTube viewer, typically high-intent (1+ hr/day), seeking niche content beyond mainstream recommendations.
AI Label Verification System	An automated system actor that classifies submitted labels against taxonomy and content signals.
Platform Admin / Trust & Safety	Internal staff who moderate flagged labels, manage taxonomy, and enforce policy.

2. Use Case Diagram

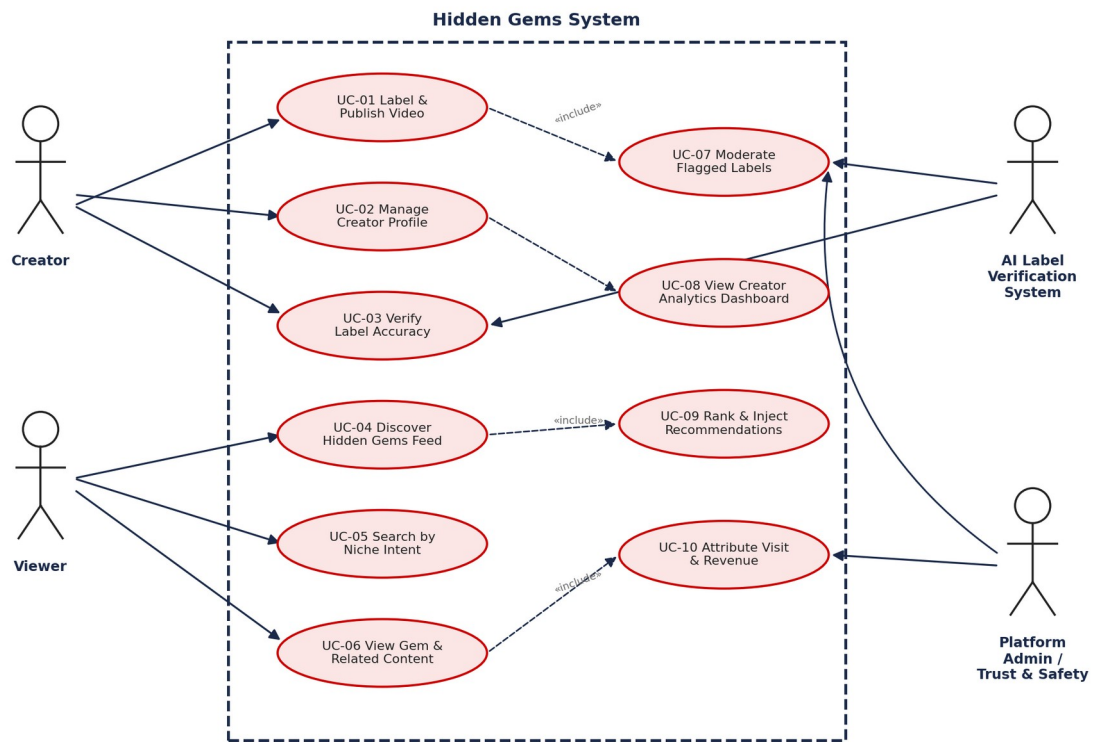


Figure 4 — Use Case Diagram

Dashed arrows denote an «include» relationship, meaning the base use case always invokes the included use case as part of its normal flow.

3. Use Case Index

ID	Name	Primary Actor	Priority
UC-01	Label & Publish Video	Creator	Must
UC-02	Manage Creator Profile	Creator	Should
UC-03	Verify Label Accuracy	AI Label Verification System	Must
UC-04	Discover Hidden Gems Feed	Viewer	Must
UC-05	Search by Niche Intent	Viewer	Must
UC-06	View Gem & Related Content	Viewer	Should
UC-07	Moderate Flagged Labels	Platform Admin / Trust & Safety	Must
UC-08	View Creator Analytics Dashboard	Creator	Should
UC-09	Rank & Inject Recommendations	AI Label Verification System*	Must
UC-10	Attribute Visit & Revenue	Platform (system)	Must

*UC-09 is executed by the Recommendation Engine, a system actor closely coupled with the AI Label Verification System in the diagram's system boundary.

4. Detailed Use Case Specifications

UC-01 — Label & Publish Video

Use Case ID	UC-01
Use Case Name	Label & Publish Video
Primary Actor	Creator
Secondary Actor(s)	AI Label Verification System
Trigger	Creator uploads a new video or edits an existing published video in YouTube Studio.
Preconditions	Creator has an active YouTube channel in good standing; Creator is authenticated in YouTube Studio
Frequency of Use	Once per video upload; occasional edits thereafter

Main (Basic) Flow:

1. Creator uploads a video file and enters title, description, and thumbnail in YouTube Studio.
2. System displays suggested niche labels based on content analysis (includes UC-03).
3. Creator selects one or more suggested labels and/or enters a custom label.
4. Creator submits the video for publishing.
5. System sends submitted labels to the AI Label Verification System for validation (includes UC-03).
6. System receives an “approved” verification result for all labels.
7. System attaches approved labels to the video and marks it Hidden Gems eligible.
8. System confirms successful publish to the Creator.

Alternate Flows:

- 3a. Creator skips labeling entirely: video publishes without Hidden Gems eligibility; system displays reach-uplift messaging encouraging labeling on a later visit.
- 6a. One or more labels are rejected: see Exception Flow E1.

Exception Flows:

- E1 — Label Rejected: System notifies the Creator with a rejection reason; video may still publish without the rejected label(s); Creator may edit and resubmit later (returns to Step 3).
- E2 — Verification Timeout: If the AI Label Verification System does not respond within the defined SLA, the system queues the video for asynchronous verification and publishes without Hidden Gems eligibility in the interim.

Postconditions:

- Success: Video is published with one or more verified labels and is eligible for the Hidden Gems discovery surface.
- Failure: Video is published without Hidden Gems eligibility; Creator is notified of any rejected labels.

Business Rules:

- A video may carry multiple labels but each label must be independently verified.
- Only creators with 1,000–500,000 subscribers are eligible for Hidden Gems surfacing (see FR-M5-02).

UC-02 — Manage Creator Profile

Use Case ID	UC-02
Use Case Name	Manage Creator Profile
Primary Actor	Creator
Secondary Actor(s)	None
Trigger	Creator navigates to their public profile settings or views their profile as rendered to viewers.
Preconditions	Creator is authenticated; Creator has at least one published video
Frequency of Use	Occasional — as needed

Main (Basic) Flow:

9. Creator navigates to channel/profile settings.
10. System displays the creator's currently active niche labels, aggregated from their labeled videos.
11. Creator reviews how their profile appears to viewers, including the Hidden Gems module.
12. Creator may unpublish or edit a video's labels, which updates the aggregated profile view (returns to UC-01, Step 3).

Alternate Flows:

- 3a. Creator has no Hidden Gems eligible content: system displays guidance on how to become eligible.

Postconditions:

- Creator profile accurately reflects the creator's current active labels and Hidden Gems content.

UC-03 — Verify Label Accuracy

Use Case ID	UC-03
Use Case Name	Verify Label Accuracy
Primary Actor	AI Label Verification System
Secondary Actor(s)	Platform Admin / Trust & Safety
Trigger	A label is submitted by a creator (from UC-01) or re-evaluated due to a taxonomy update.
Preconditions	Approved taxonomy is loaded and versioned; Video content signals (title, description, transcript/visual metadata) are available
Frequency of Use	Every label submission (high volume, near real time)

Main (Basic) Flow:

13. System receives the submitted label and associated video content signals.
14. System computes a match against the approved taxonomy and a confidence score against content signals.
15. System evaluates the confidence score against approval and rejection thresholds.
16. If the score exceeds the approval threshold, system marks the label Approved.
17. System logs the decision, confidence score, and taxonomy version used for audit.
18. System returns the verification result to the calling use case (UC-01).

Alternate Flows:

- 3a. Score falls below the rejection threshold: system marks the label Rejected and returns a rejection reason code.
- 3b. Score falls between thresholds (ambiguous): system marks the label Needs Review and routes it to UC-07.

Postconditions:

- Label status is set to Approved, Rejected, or Needs Review.
- Decision is logged with confidence score and taxonomy version for auditability (NFR-07).

Business Rules:

- Minimum acceptable precision for automated approval decisions is 90% (NFR-03).

UC-04 — Discover Hidden Gems Feed

Use Case ID	UC-04
Use Case Name	Discover Hidden Gems Feed
Primary Actor	Viewer
Secondary Actor(s)	Recommendation Engine (UC-09)
Trigger	Viewer opens the YouTube home tab.
Preconditions	Viewer is an eligible member of the target viewer segment; At least one Hidden Gems eligible video exists
Frequency of Use	Multiple times per day, per active viewer

Main (Basic) Flow:

19. Viewer opens the YouTube home tab.
20. System requests ranked Hidden Gems candidates for this viewer (includes UC-09).
21. System renders the Hidden Gems section with niche preference chips and video cards.
22. Viewer optionally selects a niche chip to filter the surface.
23. System re-renders filtered results.
24. Viewer selects a video card (continues to UC-06).

Alternate Flows:

- 4a. Viewer provides negative feedback (“not interested”) on a card: system removes the card and logs the signal for future ranking.

Postconditions:

- Viewer has viewed a personalized set of niche/mid-tier creator content on the home surface.
- Impression events are logged for attribution (UC-10).

UC-05 — Search by Niche Intent

Use Case ID	UC-05
Use Case Name	Search by Niche Intent
Primary Actor	Viewer
Secondary Actor(s)	Recommendation Engine (UC-09)
Trigger	Viewer enters a search query matching or resembling a taxonomy label.
Preconditions	Approved taxonomy is available to the search index
Frequency of Use	Multiple times per day, per active viewer

Main (Basic) Flow:

25. Viewer enters a search query.
26. System matches the query against the approved label taxonomy and offers autocomplete suggestions.
27. Viewer submits the search.
28. System retrieves and ranks Hidden Gems content matching the label (includes UC-09).
29. System renders a badged Hidden Gems section within or above standard search results.
30. Viewer selects a result (continues to UC-06).

Alternate Flows:

- 2a. No taxonomy match is found: system falls back to standard search results with no Hidden Gems section.

Postconditions:

- Viewer has been routed toward niche creator content matching stated search intent.
- Impression events are logged for attribution (UC-10).

UC-06 — View Gem & Related Content

Use Case ID	UC-06
Use Case Name	View Gem & Related Content
Primary Actor	Viewer
Secondary Actor(s)	None
Trigger	Viewer selects a Hidden Gems video card from the feed, search, or a related-content module.
Preconditions	Selected video is published and Hidden Gems eligible
Frequency of Use	Every Hidden Gems video selection

Main (Basic) Flow:

31. System renders the watch page with the “Featured in Hidden Gems” badge.
32. System renders related-content modules: “More with label” and “More gems from [creator].”
33. Viewer watches the video and may interact with related modules.
34. Viewer optionally taps the creator's name to navigate to the creator profile (continues to UC-02 viewer-facing view).

Postconditions:

- Watch-time, click, and navigation signals are logged for ranking and attribution (UC-09, UC-10).

UC-07 — Moderate Flagged Labels

Use Case ID	UC-07
Use Case Name	Moderate Flagged Labels
Primary Actor	Platform Admin / Trust & Safety
Secondary Actor(s)	AI Label Verification System
Trigger	A label is marked Needs Review by UC-03.
Preconditions	Moderator is authenticated with Trust & Safety permissions
Frequency of Use	As volume dictates; target SLA per operational policy

Main (Basic) Flow:

35. Moderator opens the moderation queue.
36. System displays the flagged label, associated video, and AI confidence score.
37. Moderator reviews the content and taxonomy context.
38. Moderator approves, rejects, or edits the label.
39. System logs the decision with moderator identity, timestamp, and rationale.
40. System notifies the creator of the final decision.

Alternate Flows:

- 4a. Moderator identifies a pattern of repeated violations by the same creator: moderator suspends the creator's labeling privileges for a stated duration.

Postconditions:

- Label reaches a final Approved or Rejected state.
- Audit log entry is created (NFR-07).

UC-08 — View Creator Analytics Dashboard

Use Case ID	UC-08
Use Case Name	View Creator Analytics Dashboard
Primary Actor	Creator
Secondary Actor(s)	None
Trigger	Creator navigates to the Hidden Gems tab within Creator Studio analytics.
Preconditions	Creator has at least one Hidden Gems eligible video with recorded activity
Frequency of Use	Weekly or as needed by the creator

Main (Basic) Flow:

41. Creator opens Creator Studio and navigates to Analytics > Hidden Gems.
42. System retrieves attributed views, CTR, subscribes, and revenue for the selected date range.
43. System renders the dashboard with trend charts and per-video breakdowns.
44. Creator adjusts the date range or filters by label.
45. System refreshes the dashboard accordingly.

Postconditions:

- Creator has visibility into Hidden Gems performance to inform labeling and content decisions.

UC-09 — Rank & Inject Recommendations

Use Case ID	UC-09
Use Case Name	Rank & Inject Recommendations
Primary Actor	Recommendation Engine (system)
Secondary Actor(s)	Viewer, Creator
Trigger	A viewer requests the Hidden Gems surface (home, search, or watch-page related content).
Preconditions	Viewer preference profile and eligible video corpus are available
Frequency of Use	Every discovery-surface request (high volume, near real time)

Main (Basic) Flow:

46. System receives a ranking request scoped to a viewer and surface context (home, search, or watch page).
47. System retrieves eligible Hidden Gems candidates from the indexed corpus.
48. System scores candidates using label affinity, watch history, and freshness signals.
49. System filters out any candidate exceeding the creator subscriber ceiling.
50. System returns a ranked list to the calling surface (UC-04, UC-05, or UC-06).

Alternate Flows:

- 3a. Insufficient candidates exist for the viewer's preferences: system broadens ranking to adjacent labels to avoid an empty surface.

Postconditions:

- A relevance-optimized ranked list is delivered within the p95 < 400ms latency guardrail (NFR-02).

Business Rules:

- Recommendation relevance must meet or exceed an 85% relevance target (NFR-04).

UC-10 — Attribute Visit & Revenue

Use Case ID	UC-10
Use Case Name	Attribute Visit & Revenue
Primary Actor	Platform (system)
Secondary Actor(s)	Creator
Trigger	A viewer completes an action (visit, subscribe, or revenue event) following a Hidden Gems impression.
Preconditions	An impression event exists for the viewer/video pair within the attribution window
Frequency of Use	Continuous, event-driven

Main (Basic) Flow:

51. System detects a downstream event (profile visit, subscribe, ad impression, membership, or Super Chat).
52. System checks for a qualifying Hidden Gems impression within the attribution window.
53. System tags the event with source = hidden_gems and links it to the originating video/label.
54. System aggregates attributed events into daily reporting (North Star Metric and supporting KPIs).
55. System makes attributed metrics available to the Creator Analytics Dashboard (UC-08) and internal reporting.

Alternate Flows:

- 2a. No qualifying impression exists within the window: event is recorded normally without Hidden Gems attribution.

Postconditions:

- Attributed metrics accurately reflect Hidden Gems' incremental contribution to creator and platform outcomes.